Chapter 3: Minimum potential energy

Solutions to the exercises

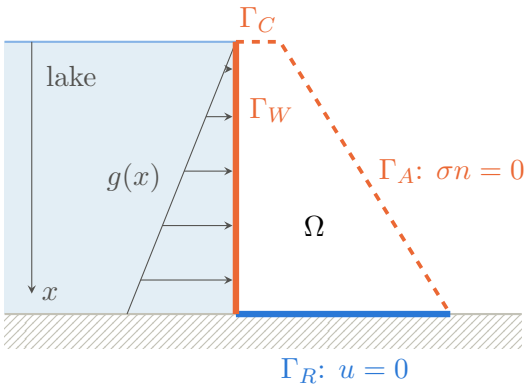
Exercise 2: Linear elasticity, a dam

Linear elasticity has $O(A) = 2$, $s = 1$, $m = 0$: the displacement ($c = 0$) is an essential, the traction σn (first derivatives, $c = 1$) a natural boundary condition.

1.

- Γ_R (rock bed): **Dirichlet**, the dam is fixed to the rock: $u = 0$.
- Γ_W (water side): **Neumann**, the water pressure acts against the outward normal and pushes the dam downstream: $\sigma n = T = -g(x)\, n$, with $n = (-1, 0)$ on the vertical face.
- Γ_C (crest) and Γ_A (air side): **homogeneous Neumann**, free surfaces: $\sigma n = 0$.

In $g(x) = \rho\, g\, x$, x is the depth below the water surface: with the vertical coordinate y and the water surface at $y = H$, $x = H - y$.



2. The test functions vanish where u is prescribed, $v = 0$ on Γ_R; there the traction (the unknown reaction force of the rock) drops out of the boundary integral. On Γ_W the traction is $T = -g(x)\, n$, on Γ_C and Γ_A it is zero:

$$\int_{\partial\Omega} T \cdot v \, ds = -\int_{\Gamma_W} g(x)\, n \cdot v \, ds = \int_{\Gamma_W} g(x)\, v_1 \, ds.$$

The variational form of the dam: find u with $u = 0$ on Γ_R such that

$$\int_\Omega \sigma(u) : \varepsilon(v) \, dx = \int_\Omega f \cdot v \, dx + \int_{\Gamma_W} g(x)\, v_1 \, ds \qquad \text{for all } v \text{ with } v = 0 \text{ on } \Gamma_R,$$

with the own weight of the concrete $f = (0, -\rho_c\, g)$ as body force ($\rho_c \approx 2400\,\mathrm{kg/m^3}$, water $\rho = 1000\,\mathrm{kg/m^3}$). The essential condition on Γ_R is built into the function space (`DirichletBC`), the inhomogeneous natural condition on Γ_W appears as a boundary integral, and the homogeneous ones on Γ_C and Γ_A by omission.

Strong form for comparison: $-\nabla \cdot \sigma(u) = f$ in Ω, $u = 0$ on Γ_R, $\sigma n = -g(x)\, n$ on Γ_W, $\sigma n = 0$ on $\Gamma_C \cup \Gamma_A$.